

EE 581: Applied Statistics for Data and Signals

Course Code	EE 581
Course Name	Applied Statistics for Data and Signals
L-T-P-C	3-0-2-4
Prerequisites	IC 210/ IC252 Probability, statistics, and random variables or equivalent.
Intended for	MTech/MTech (R)/ PhD students of the SCEE
Mutual Exclusion	EE534, CS511, MA 511, MA521, MA524, MA568, MA 601, or equivalent.
Approval	54th BoA (May 2024)

Course Contents

- **Analysis:** The Real Number System, Euclidean Spaces, Metric Spaces, Closed and open sets. (6 Hours)
- **Module II:** Numerical sequences and series, convergence of sequences of numbers, Limits, Continuity of functions, Derivatives of functions, Integration (Riemann integration, Lebesgue integration, Riemann-Stieltjes, Lebesgue-Stieltjes). (8 Hours)
- **Probability Theory:** The axioms of probability theory, Independence and conditional probability, Random variables and their distribution, Expectation, Conditional distribution, Convergence of sequences of random variables, Laws of large numbers and Central limit theorem. General concepts of stochastic processes. (11 Hours)
- **Data driven/ Statistical detection and Estimation:** Hypothesis testing, Optimal detectors, minimum variance unbiased estimation, mean squared estimation; filtering, and prediction, and Kalman filtering, parametric/ nonparametric distribution estimation, data driven detection and estimation. (11 Hours)
- **Basics of Queueing theory:** Introduction to Markov chains, Kendall's notation for queueing systems, M/M/1, M/M/m, M/M/infinity, M/M/m/m, M/G/1, G/G/1 queueing systems and applications. (6 Hours)

Laboratory Modules

- The two-hours of lab session per week will enhance the understanding of the concepts taught in the class. The lab will cover the concepts including sets, sequence convergence, random variables, central limit theorem, detection, and estimation techniques for statistical and data driven scenarios.

Text Books

1. Hajek, B., **Random Processes for Engineers**, Cambridge University Press, 2015.
2. S. M. Kay, **Fundamentals of Statistical Signal Processing, Vol - 1 & 2**, Pearson, 20210.

References

1. Deisenroth, M. P., Faisal, A. A, Ong, C. S., **Mathematics for Machine Learning**, Cambridge University Press, 2020.
2. J. G. Proakis and M. Salehi, **Fundamentals of Communication Systems**, Prentice Hall, 2004.
3. Harry L. Van Trees, Kristine L. Bell, Zhi Tian, **Detection, Estimation, and Modulation Theory: Detection, Estimation, and Linear Modulation Theory**, Wiley, 2013.
4. Dimitri P. Bertsekas, and Robert G. Gallager, **Data Networks**, 2nd Edition, Pearson Publication